

The AI boom and geopolitics are rewiring Asia's oceans

New cables between data centres are avoiding China and chokepoints

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illustration: ravenjiang

“The seabed is a battlefield,” Australia’s defence minister told a room full of admirals and generals in Singapore at the end of May. Richard Marles, citing several subsea cables that have been cut in the Baltic Sea and around Taiwan in recent years, joined 16 of his counterparts in announcing plans to protect the submarine tendrils of the digital world: the nearly 700 communications cables which mostly lie exposed on the floor of the world’s oceans.

Governments and armed forces in Asia have only recently awoken to the importance of subsea cables. Some of their fears of subterfuge may be overblown; to date, no conclusive evidence has been shared to suggest that the cuts highlighted by Mr Marles are sabotage. But they are right about the vulnerability of these arteries of commerce. And the private firms which build and operate almost all of the world’s subsea cables are not waiting for governments to better secure them. They are increasingly taking steps on their own to avoid Asia’s most contested waters.

Asia and Australia are now connected to Europe by fibre-optic cables which tend to hug the coastlines of the Asian continent before heading up the Red Sea. But a combination of the ai boom and geopolitics is rerouting cable traffic across the Indian and Pacific oceans. This new

geography avoids chokepoints like the Strait of Malacca and contested waters like the South China Sea. Much of it avoids South-East Asia entirely, running from the Middle East and India to Australia and then onward through the Pacific Islands to America.

The first cable to run the new route was laid in 2022 between Oman and Australia, with spurs to the Anglo-American military base at Diego Garcia and the Cocos Islands, a tiny Australian territory in the Indian Ocean. Then, last year, Google announced that Christmas Island, another Australian territory in the Indian Ocean, would become a hub for a new cable network between Australia and the Middle East. Fibre will run from Oman via the Maldives to Christmas Island, and then onward to Australia. Meta's \$10bn Project Waterworth, a global cable network still in development, looks set to follow a similar course in the Indian Ocean.

The first shift remapping cable routes is a change in who pays for them. Subsea cables are expensive. To defray the cost, for most of the last few decades big national telecoms firms would form consortia to build them. Back in 1999, one of the first big fibre-optic cables between Europe and Asia to come online, known as sea-me-we 3, cost \$1.3bn and had 92 consortium partners. Financing and planning a cable among so many firms tended to increase costs and delay laying it. Once it was funded, the number of partners involved pulled the route close to the Asian continent where the bulk of the customers were located.

But the ai boom is scrambling the economics of the subsea cable business and changing its geography. Over the past ten years, internet giants have begun to finance and build cables single-handedly. That has simplified the fundraising and planning process, and cut the lead-time for new cable projects. Google invested in its first cable in 2008. It has since funded at least 34 more, 18 of which it owns without partners. Increasingly, firms like Meta, Google and Microsoft are building cables not to connect population centres but to connect their data centres.

And build them they are. By one estimate, the next four years will see an average of \$4bn a year in new cable investment, the bulk of it by so-called hyperscalers seeking to win the ai race. While satellite internet service from firms such as Starlink is getting cheaper, it is still orders of magnitude more expensive to beam each gigabyte of data into space than it is to push light down a cable, and is likely to be so for many years yet. As a result, subsea cables still carry 99% of the world's intercontinental internet traffic.

As the subsea cable market consolidates vertically, it is expanding geographically. Unshackled from the need to remain close to population centres, ships are laying subsea cables across the open ocean more than ever before. The new routes have been drawn up to avoid seabed governed either by China or by governments that might seek to extract payment for laying or repairing a cable across a chokepoint, such as the Indonesian straits.

Geopolitical risk has become particularly acute in the South China Sea, where China has yet to effect full control on the surface but exercises de facto sovereignty over the seabed. Under

international law, states are not supposed to interfere with repairs to cables outside their territorial sea. But repairs to any cables within China's "nine-dash" line, which stretches over a thousand kilometres from China's coasts (and which it claims as the extent of its waters) require approval from officials in Beijing.

Cables transiting through chokepoints like the Strait of Malacca run similar risks, says Samuel Bashfield, who studies subsea cables at La Trobe University in Australia. Constantly shifting rules set by littoral countries like Malaysia and Indonesia are designed to extract value from cable operations through measures like requiring the use of local ships. These can be expensive annoyances. But recent musings by Indonesia's cash-strapped president and finance minister about how the country could make money from its position astride some of the world's great sea-lanes suggest that more aggressive measures could be coming.

To avoid these risky shoals, more and more internet traffic is simply going around them. Google and Meta's new networks run instead from the Middle East through Australia and onwards to Japan and South Korea or America. In the Pacific, cables increasingly use Guam as a hub to connect American allies in Asia. These new routes are part of an increasingly bifurcated internet infrastructure beneath the waves: no new cables between America and China have been approved since Barack Obama was in office. ■

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